

Create a C program to manage student marks for multiple departments. The program must use a three-dimensional array to store the data and perform the following tasks:

- **Data Structure:** The array marks[5][3][5] should be used to store marks for 5 departments, with 3 students per department, each taking 5 subjects.
- **User Input:** The program must first prompt the user to input marks for every student in every subject across all 5 departments.
- **Total and Average Calculation:** After data entry is complete, the program should ask the user to select a specific department. For the chosen department, it must then calculate and display the total marks and average marks for each student.
- **Indexing:** The program should handle the user-friendly input (e.g., department 1, student 1) by correctly converting it to the 0-based indexing required for the C array.

CODE:

```
int main(){
    float marks[5][3][5];
    for(int i =0; i < 5 ; i++){
        printf("\n###Enter the marks for Department %d ### \n", i + 1 );
        for(int j = 0 ; j < 3 ; j++){
            printf("\n###Enter the marks for Student %d ###\n\n", j + 1);
            for(int k = 0; k < 5; k++){
                printf("Enter the marks for Subject %d : ",k + 1);
                scanf("%f", &marks[i][j][k]);
            }
        }
    }
    int dept;
    float total[3]= {0};
    float avg[3] ={0};
    printf("\nSelect the Department (1,2,3,4,5) : ");
    scanf("%d", &dept);
    int i = dept;
    for(int j = 0 ; j < 3 ; j++){
        float sum = 0;
        for(int k = 0; k < 5; k++){
            sum += marks[i][j][k];
        }
        total[j] = sum;
        avg[j] = sum/5;
    }

    printf("Result for Department %d\n", dept);
    for(int x = 0 ; x < 3; x++ ){
        printf("Student %d \n", x + 1);
        printf("Total Marks %f \t", total[x]);
        printf("Average Marks %f\n", avg[x]);
    }
}
```

OUTPUT :

```
###Enter the marks for Department 1 ###
```

```
###Enter the marks for Student 1 ###
```

```
Enter the marks for Subject 1 : 86  
Enter the marks for Subject 2 : 79  
Enter the marks for Subject 3 : 47  
Enter the marks for Subject 4 : 86  
Enter the marks for Subject 5 : 86
```

```
###Enter the marks for Student 2 ###
```

```
Enter the marks for Subject 1 : 90  
Enter the marks for Subject 2 : 86  
Enter the marks for Subject 3 : 75  
Enter the marks for Subject 4 : 82  
Enter the marks for Subject 5 : 75
```

```
###Enter the marks for Student 3 ###
```

```
Enter the marks for Subject 1 : 93  
Enter the marks for Subject 2 : 75  
Enter the marks for Subject 3 : 62  
Enter the marks for Subject 4 : 72  
Enter the marks for Subject 5 : 86
```

```
###Enter the marks for Department 2 ###
```

```
###Enter the marks for Student 1 ###
```

```
Enter the marks for Subject 1 : 91  
Enter the marks for Subject 2 : 85  
Enter the marks for Subject 3 : 79  
Enter the marks for Subject 4 : 69  
Enter the marks for Subject 5 : 84
```

```
###Enter the marks for Student 2 ###
```

```
Enter the marks for Subject 1 : 83  
Enter the marks for Subject 2 : 92  
Enter the marks for Subject 3 : 86  
Enter the marks for Subject 4 : 91  
Enter the marks for Subject 5 : 76
```

```
###Enter the marks for Student 3 ###
```

```
Enter the marks for Subject 1 : 78  
Enter the marks for Subject 2 : 98  
Enter the marks for Subject 3 : 78  
Enter the marks for Subject 4 : 84  
Enter the marks for Subject 5 : 59
```

```
###Enter the marks for Department 3 ###
```

```
###Enter the marks for Student 1 ###
```

```
Enter the marks for Subject 1 : 50  
Enter the marks for Subject 2 : 80  
Enter the marks for Subject 3 : 89  
Enter the marks for Subject 4 : 72  
Enter the marks for Subject 5 : 60
```

```
###Enter the marks for Student 2 ###
```

```
Enter the marks for Subject 1 : 74  
Enter the marks for Subject 2 : 68  
Enter the marks for Subject 3 : 45  
Enter the marks for Subject 4 : 79  
Enter the marks for Subject 5 : 52
```

```
###Enter the marks for Student 3 ###
```

```
Enter the marks for Subject 1 : 50  
Enter the marks for Subject 2 : 62  
Enter the marks for Subject 3 : 84  
Enter the marks for Subject 4 : 73  
Enter the marks for Subject 5 : 79
```

###Enter the marks for Department 4 ###

###Enter the marks for Student 1 ###

Enter the marks for Subject 1 : 82
Enter the marks for Subject 2 : 50
Enter the marks for Subject 3 : 81
Enter the marks for Subject 4 : 75
Enter the marks for Subject 5 : 92

###Enter the marks for Student 2 ###

Enter the marks for Subject 1 : 86
Enter the marks for Subject 2 : 76
Enter the marks for Subject 3 : 74
Enter the marks for Subject 4 : 69
Enter the marks for Subject 5 : 63

###Enter the marks for Student 3 ###

Enter the marks for Subject 1 : 92
Enter the marks for Subject 2 : 86
Enter the marks for Subject 3 : 73
Enter the marks for Subject 4 : 69
Enter the marks for Subject 5 : 72

###Enter the marks for Department 5 ###

###Enter the marks for Student 1 ###

Enter the marks for Subject 1 : 65
Enter the marks for Subject 2 : 50
Enter the marks for Subject 3 : 59
Enter the marks for Subject 4 : 86
Enter the marks for Subject 5 : 79

###Enter the marks for Student 2 ###

Enter the marks for Subject 1 : 72
Enter the marks for Subject 2 : 63
Enter the marks for Subject 3 : 72
Enter the marks for Subject 4 : 79
Enter the marks for Subject 5 : 75

###Enter the marks for Student 3 ###

Enter the marks for Subject 1 : 69
Enter the marks for Subject 2 : 62
Enter the marks for Subject 3 : 92
Enter the marks for Subject 4 : 73
Enter the marks for Subject 5 : 74

Select the Department (1,2,3,4,5) : 3

Result for Department 3

Student 1

Total Marks 380.000000 Average Marks 76.000000

Student 2

Total Marks 368.000000 Average Marks 73.599998

Student 3

Total Marks 392.000000 Average Marks 78.400002